DATA ENGINEERING

Stream Processing Visual Guide

Data in motion —
from batch thinking to real-time thinking.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Batch vs Stream

Batch

hourly / daily jobs

- 'What happened yesterday?'
- Complete data, simple logic
- High latency by design

Stream

milliseconds

- 'What's happening NOW?'
- Infinite data, windowed logic
- Low latency, harder ops

Lambda: batch + stream paths (complex). Kappa: stream-only, reprocess from the log (simpler).

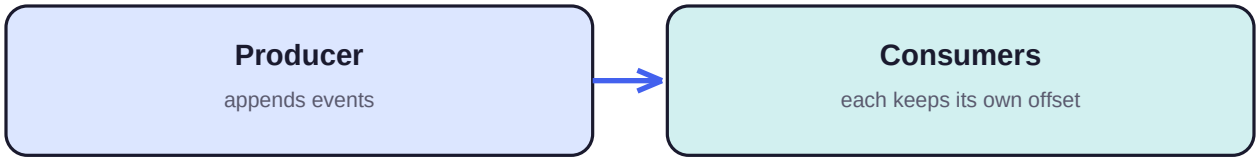
STEP 2/6

The Log: Foundation of Streaming

An append-only, replayable sequence — Kafka's big idea.

Immutable, ordered, partitioned. Consumers track their own offset — replay anytime.

Retention (days–forever) decouples producers from consumers in TIME as well as space.



STEP 3/6

Windows: Taming Infinity

You can't aggregate 'all data ever' — slice time into windows.

Tumbling

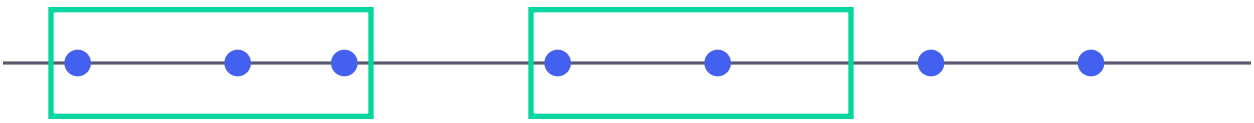
Fixed, non-overlapping: every 5 min. Simple counts.

Sliding

Overlapping: every 1 min over last 5. Smoother.

Session

Gaps define boundaries: user activity bursts.



tumbling windows

STEP 4/6

Event Time vs Processing Time

When it happened vs when you saw it — they differ.

Event time: the timestamp IN the event (phone's clock).

Processing time: when your system received it (server's clock).

Late data: events arriving after their window closed. Watermarks declare 'we've seen everything up to T' — then fire the window, with allowed lateness for stragglers.

- Use event time for correctness (billing, analytics).
- Use processing time only when approximate is fine (dashboards).
- Exactly-once: idempotent sinks + transactional writes (Kafka transactions, Flink checkpoints).

STEP 5/6

Exactly-Once Semantics

The holy grail: no duplicates, no loss.

At-most-once

May lose. Fire and forget.

At-least-once

May duplicate. Retry until acked.

Exactly-once

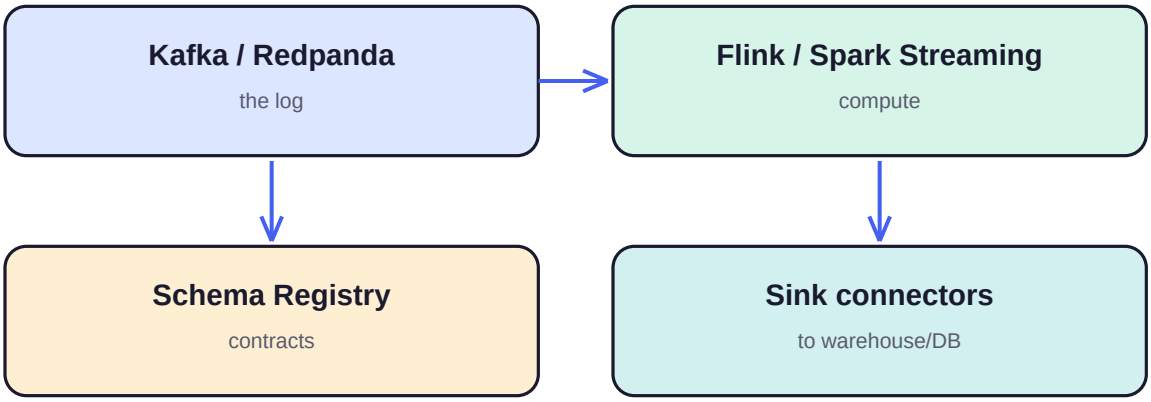
Idempotent consumers + transactional commits.

Recipe: deterministic IDs + idempotent sinks (upserts by key) + checkpointed offsets.

STEP 6/6

The Streaming Stack

Pieces that fit together.



Schema Registry prevents 'field renamed, pipeline broke at 2am' — compatibility checked at publish.

STEP ✓I✓

Key Takeaways

Stream Processing — Remember

- The log (Kafka) is the foundation: immutable, ordered, replayable.
- Windows slice infinity: tumbling, sliding, session — pick per question.
- Event time = correctness; watermarks handle lateness.
- Exactly-once = idempotent sinks + transactional offsets.
- Schema Registry keeps producers and consumers in contract.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev